

# SQL vs KQL: Basic Comparison

Reference table: Telemetry | Columns: Timestamp, DeviceId, Temperature, Status

| Operation                         | SQL                                                                                                      | KQL                                                                                   |
|-----------------------------------|----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>1. Select / limit rows</b>     | SELECT TOP 10 * FROM Telemetry;                                                                          | Telemetry<br>  take 10                                                                |
| <b>2. Select specific columns</b> | SELECT DeviceId, Temperature FROM Telemetry;                                                             | Telemetry<br>  project DeviceId, Temperature                                          |
| <b>3. Filter rows</b>             | SELECT * FROM Telemetry<br>WHERE Temperature > 80 AND Status = 'Active';                                 | Telemetry<br>  where Temperature > 80 and Status == "Active"                          |
| <b>4. Sort</b>                    | SELECT * FROM Telemetry ORDER BY Timestamp DESC;                                                         | Telemetry<br>  order by Timestamp desc                                                |
| <b>5. Computed columns</b>        | SELECT DeviceId, Temperature * 9/5 + 32 AS TempF<br>FROM Telemetry;                                      | Telemetry<br>  extend TempF = Temperature * 9.0 / 5 + 32<br>  project DeviceId, TempF |
| <b>6. Aggregation (GROUP BY)</b>  | SELECT DeviceId, AVG(Temperature) AS AvgTemp, COUNT(*) AS Events<br>FROM Telemetry<br>GROUP BY DeviceId; | Telemetry<br>  summarize AvgTemp = avg(Temperature), Events = count() by DeviceId     |
| <b>7. HAVING</b>                  | SELECT DeviceId, COUNT(*) AS Events<br>FROM Telemetry<br>GROUP BY DeviceId<br>HAVING COUNT(*) > 100;     | Telemetry<br>  summarize Events = count() by DeviceId<br>  where Events > 100         |

| Operation                    | SQL                                                                                                                                                                                   | KQL                                                                                                  |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| <b>8. Time filtering</b>     | SELECT * FROM Telemetry<br>WHERE Timestamp >= DATEADD(hour, -1, GETUTCDATE());                                                                                                        | Telemetry<br>  where Timestamp > ago(1h)                                                             |
| <b>9. Time bucketing</b>     | SELECT DATEADD(minute, DATEDIFF(minute, 0, Timestamp), 0) AS Bucket,<br>AVG(Temperature) AS AvgTemp<br>FROM Telemetry<br>GROUP BY DATEADD(minute, DATEDIFF(minute, 0, Timestamp), 0); | Telemetry<br>  summarize AvgTemp = avg(Temperature) by bin(Timestamp, 1m)                            |
| <b>10. Distinct values</b>   | SELECT DISTINCT DeviceId FROM Telemetry;                                                                                                                                              | Telemetry<br>  distinct DeviceId                                                                     |
| <b>11. String matching</b>   | SELECT * FROM Telemetry WHERE DeviceId LIKE 'sensor%';                                                                                                                                | Telemetry<br>  where DeviceId startswith "sensor"<br>-- also: contains, has, endswith, matches regex |
| <b>12. Join</b>              | SELECT t.DeviceId, d.Location, t.Temperature<br>FROM Telemetry t<br>INNER JOIN Devices d ON t.DeviceId = d.DeviceId;                                                                  | Telemetry<br>  join kind=inner Devices on DeviceId<br>  project DeviceId, Location, Temperature      |
| <b>13. Conditional logic</b> | SELECT DeviceId,<br>CASE WHEN Temperature > 80 THEN 'Hot' ELSE 'Normal' END AS Level<br>FROM Telemetry;                                                                               | Telemetry<br>  extend Level = iff(Temperature > 80, "Hot", "Normal")<br>  project DeviceId, Level    |
| <b>14. Top N per group</b>   | SELECT TOP 5 DeviceId, MAX(Temperature) AS MaxTemp<br>FROM Telemetry<br>GROUP BY DeviceId<br>ORDER BY MaxTemp DESC;                                                                   | Telemetry<br>  summarize MaxTemp = max(Temperature) by DeviceId<br>  top 5 by MaxTemp desc           |